Bitcoin Backtesting Engine

Framing:

You are working with an analyst who is developing a bitcoin trading strategy. The strategy is as follows:

- They buy 1 unit of bitcoin at the first second of a specific hour and sell it at the last second of that hour.
- They repeat this buy/sell every day, always using the same hour for trades. For example, if they choose 15:00 (3:00 PM), they will buy at 15:00:00 and sell at 15:59:59 every day.
- After the initial purchase of 1 unit, all profits/losses are fully reinvested in subsequent buys/sells.

The analyst wants to answer the following questions:

1. Which hour of the day had the biggest returns?
2. Which hour of the day had the lowest maximum losses?

The analyst would like to be able to easily repeat this analysis for different hours and days of entering the market.

Requirements:

Using a dataset from Kaggle (https://www.kaggle.com/datasets/tzelal/binance-bitcoin-dataset-1s-timeframe-p2) , you need to submit a bash script that:

- Starts an instance of a relational database (Postgres or MySQL).
- Loads the necessary data into the database.
- Executes dbt-build command(s) to create the required dbt models.
- Runs a final SQL query on the dbt models to answer the prompt questions.

The necessary files should be submitted in an archived folder or via a link to a private GitHub repository. Do **not** include the dataset in the submission; assume the data is in the root directory when the bash script is run. A visualization of the final result is nice to have but not required.

Expectations:

You have 7 calendar days to complete the assignment. Based on previous candidates, this task typically takes between 2 to 6 hours. When evaluating your submission, we will consider:

- The completeness of the solution and your ability to make well-founded assumptions when requirements are ambiguous.
- The quality of your code in terms of verification, testing, and collaborative programming.

- Your approach to designing the transformation layer.

Note: The focus of the assignment is on the transformation portion, so less emphasis will be placed on the data-loading process.